FIG. 1 MAIN VIEW – UNCREWED AIRCRAFT PLATFORM (UCAS)

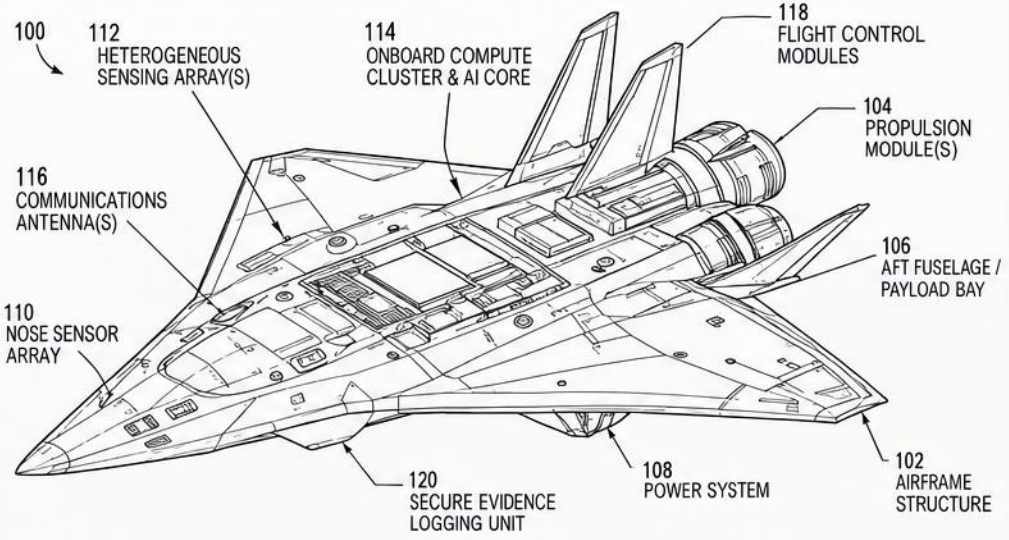
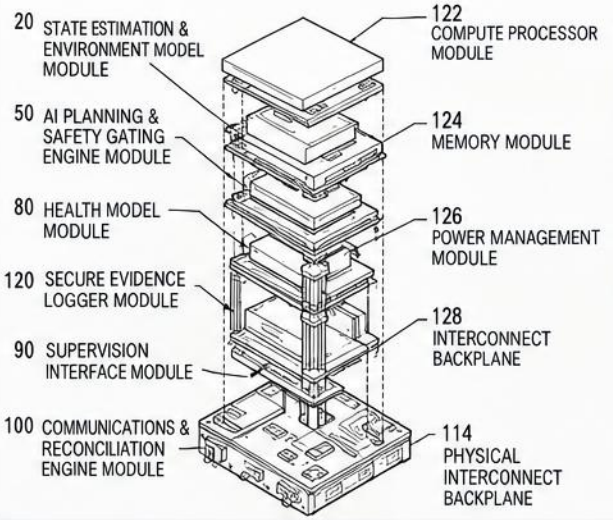
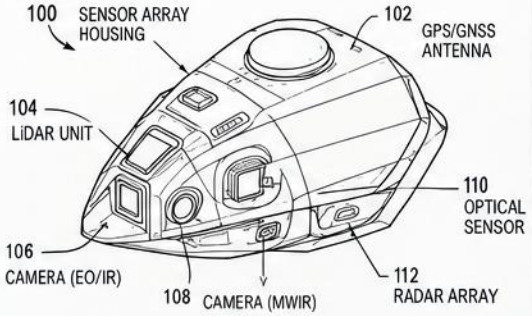


FIG. 2 EXPLODED ISOMETRIC VIEW OF ONBOARD COMPUTE CLUSTER & AI CORE (Detail of 114 from Fig. 1)



ONBOARD COMPUTE CLUSTER (TYPICAL)	
Length	12.4 m
Width	8.7 m
Height	2.6 m
Mass (max)	4,500 kg
Power	12 kW
Endurance	12+ hrs
Range	2,000+ km

FIG. 3 DETAILED VIEW OF SENSOR ARRAY (100)



SENSOR SUITE (TYPICAL))			
102	GPS/GNSS	114	IMU
104	LiDAR	116	Air Data
106	EO/IR Cameras	118	Vision
108	MWIR Camera	120	Finish/Nav
110	Optical Sensor		
112	Radar / SAR		

FIG. 4 AI PLANNING WITH DETERMINISTIC SAFETY GATING

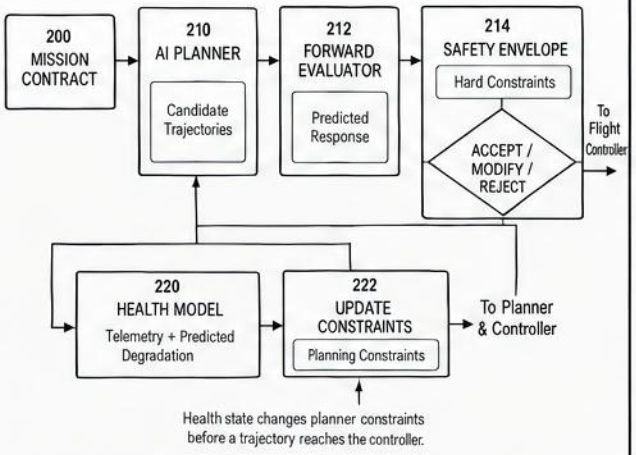
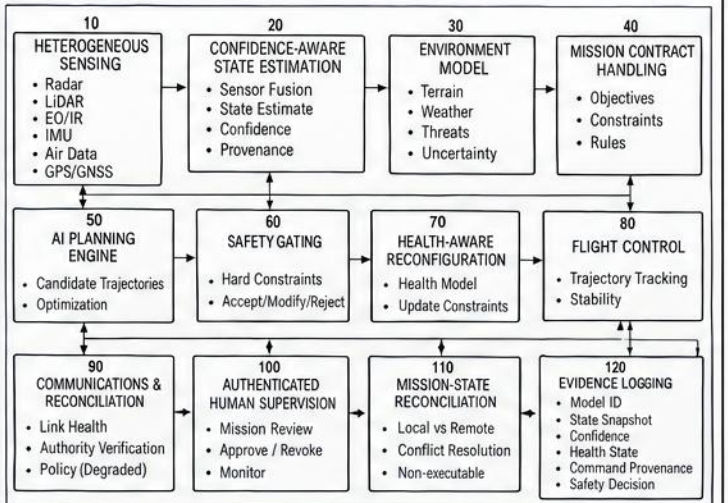


FIG. 5 SYSTEM OVERVIEW – AUTONOMY, SAFETY, RESILIENCE, AND EVIDENCE



FIGURES 1–5 | UNCREWED AIRCRAFT PLATFORM WITH INTEGRATED SENSING, AI PLANNING, SAFETY GATING, HEALTH-AWARE RECONFIGURATION, AUTHENTICATED SUPERVISION, MISSION-STATE RECONCILIATION, AND EVIDENCE LOGGING

FIG. 1 (MAIN VIEW)

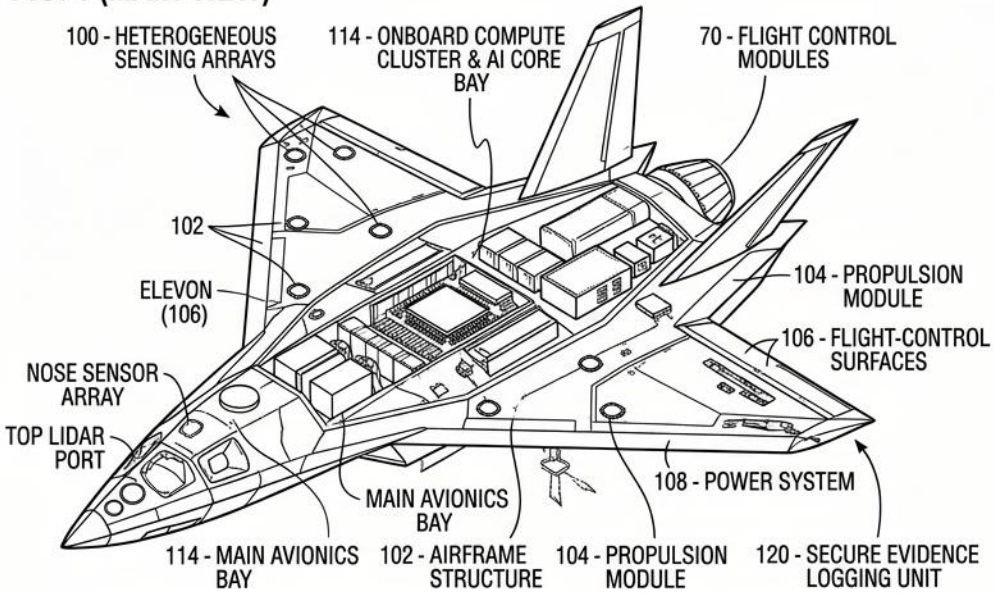


FIG. 2: EXPLODED ISOMETRIC VIEW OF ONBOARD COMPUTE CLUSTER & AI CORE

(Detail of 114 from Fig. 1)

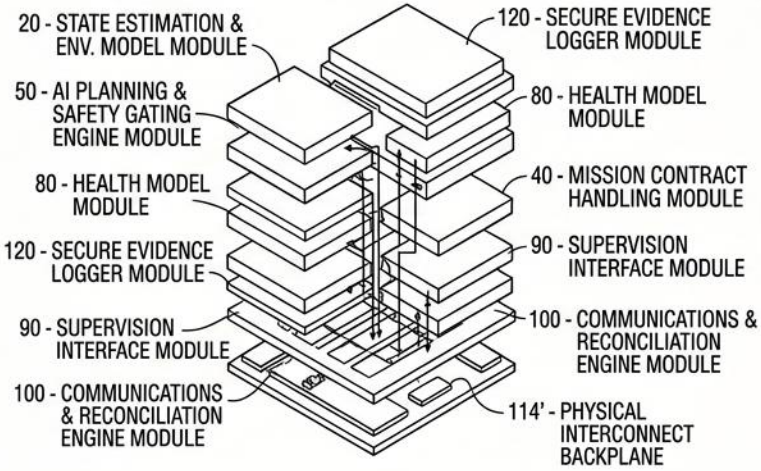


FIG. 3: DETAILED VIEW OF SENSOR ARRAY

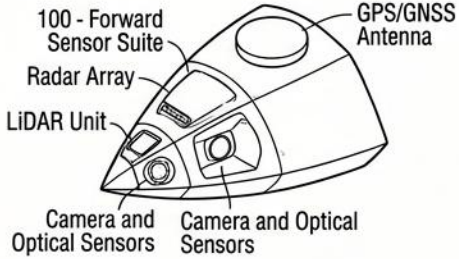


FIG. 4: DETAILED VIEW OF HEALTH MODEL TELEMETRY

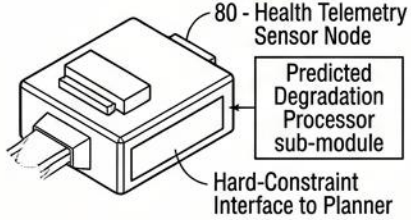
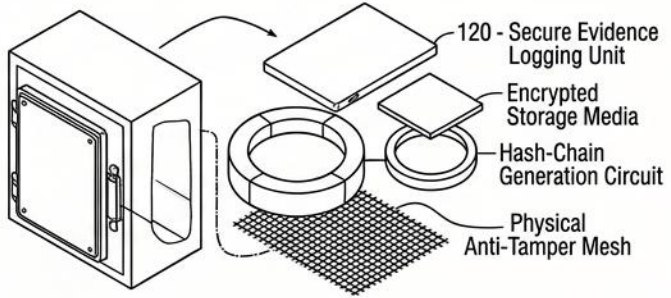
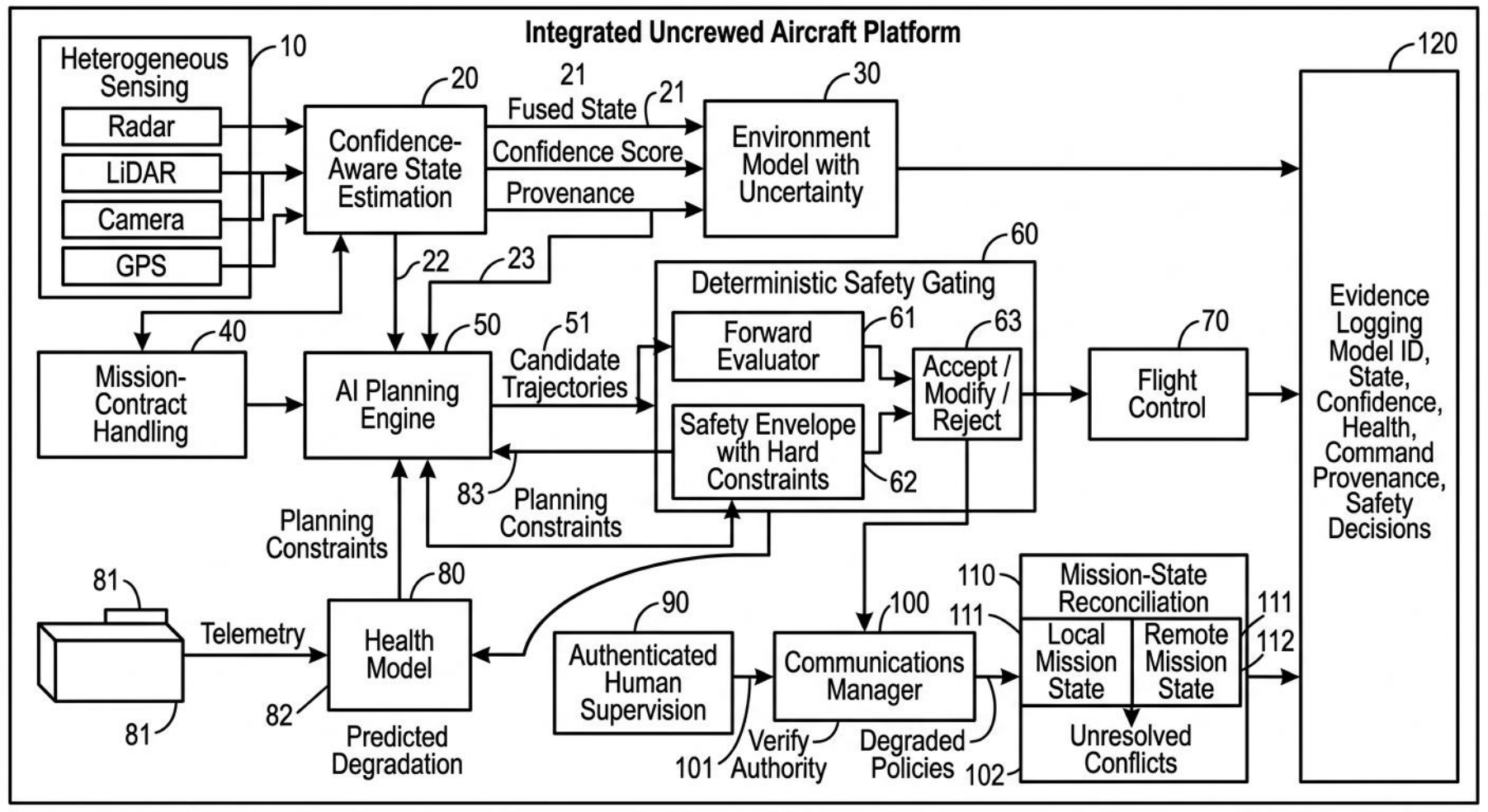


FIG. 5: EXPLODED VIEW OF SECURE EVIDENCE LOGGING UNIT




Integrated Uncrewed Aircraft Platform
Heterogeneous Sensing
10
Radar
LiDAR
Camera
GPS
Confidence-Aware State Estimation
20
21
Fused State
21
Confidence Score
Provenance
Environment Model with Uncertainty
30
120
Evidence Logging Model ID, State, Confidence, Health, Command Provenance, Safety Decisions
22
23
Deterministic Safety Gating
60
Forward Evaluator
61
Accept / Modify / Reject
63
Safety Envelope with Hard Constraints
62
Flight Control
70
Mission-Contract Handling
40
AI Planning Engine
50
51
Candidate Trajectories
83
Planning Constraints
Planning Constraints
81
81
Telemetry
Health Model
80
82
Predicted Degradation
Authenticated Human Supervision
90
101
Verify Authority
Communications Manager
100
Degraded Policies
Mission-State Reconciliation
110
111
Local Mission State
Remote Mission State
111
112
Unresolved Conflicts
102